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Energy-aware spike budgeting for continual learning in spiking neural networks for neuromorphic vision

Anika Tabassum Meem¹ , Muntasir Hossain Nadid¹ and Md Zesun Ahmed Mia^{1,2,*} ¹ Department of Electrical and Electronics Engineering, University of Liberal Arts Bangladesh, Dhaka, Bangladesh² Department of Electrical Engineering, Pennsylvania State University, University Park, PA 16802, United States of America

* Author to whom any correspondence should be addressed.

E-mail: anika.tabassum.eee@ulab.edu.bd, muntasir.hossain.eee@ulab.edu.bd and zesun.ahmed@psu.edu**Keywords:** spiking neural networks, continual learning, neuromorphic computing, event-based vision, spike-rate control

Abstract

Neuromorphic vision systems based on spiking neural networks (SNNs) offer event-driven sparse computation for event-based and frame-based cameras, yet catastrophic forgetting remains a critical barrier to deployment in continually evolving environments. Existing continual learning methods, developed primarily for artificial neural networks, seldom jointly optimize accuracy and activity-dependent spike cost, with particularly limited exploration on event-based datasets. We propose an energy-aware spike budgeting framework for continual SNN learning that integrates experience replay, learnable leaky integrate-and-fire neuron parameters, and an adaptive spike-budget controller to enforce dataset-specific spike-activity constraints during training. Our approach exhibits modality-dependent behavior: on frame-based datasets (MNIST, CIFAR-10), spike budgeting acts as a sparsity-inducing regularizer, improving accuracy while reducing spike rates by up to 47%; on event-based datasets (DVS-Gesture, N-MNIST, CIFAR-10-DVS), controlled budget relaxation enables accuracy gains up to 17.45% points with minimal computational overhead. Across five benchmarks spanning both modalities, our method demonstrates consistent performance improvements while keeping spike activity under explicit control, supporting practical continual learning under an activity-driven synaptic-event proxy rather than a direct hardware-energy measurement.

1. Introduction

Neuromorphic computing has emerged as a paradigm shift in artificial intelligence, drawing inspiration from the sparse event-driven signaling and temporal dynamics of the biological brain. Recent research has delved deeper into bio-plausible architectures, ranging from astromorphic transformers [1] to bio-inspired sequence processing mechanisms, all aiming to bridge the gap between biological and artificial intelligence [2–4]. These systems, particularly those based on spiking neural networks (SNNs), promise activity-sparse perception by leveraging discrete, event-driven computations compatible with event-based dynamic vision sensor (DVS) cameras [5]. However, deploying these systems in continually changing environments remains challenging due to catastrophic forgetting—the abrupt loss of previously learned knowledge when new tasks are introduced—a central failure mode in continual learning [6].

The continual learning literature has developed numerous strategies to mitigate forgetting in artificial neural networks (ANNs). Regularization-based approaches, such as elastic weight consolidation (EWC) [7] and Synaptic Intelligence [8], selectively consolidate important parameters to preserve past knowledge. Architectural methods like progressive neural networks [9] and PackNet [10] allocate dedicated capacity for new tasks. Memory-based approaches, including Experience Replay [11], gradient episodic memory (GEM) [12], and online retrieval methods [13, 14], explicitly store and rehearse past examples. While effective for dense, synchronous ANNs, these methods rarely consider the unique constraints and opportunities of event-driven, sparse spiking architectures.

Extending these principles to SNNs introduces additional complexity. Recent work in spiking continual learning has explored Bayesian approaches [15] that leverage probabilistic inference for uncertainty-aware learning, context-aware mechanisms [16] that adaptively modulate network resources based on task similarity, and on-chip learning frameworks [17, 18] that enable real-time adaptation on neuromorphic hardware. Domain-specific applications have also been investigated, including the ICONS'25 semi-supervised neuromorphic cybersecurity pipeline [19] and temporal sequence processing. However, these methods either assume sparsity is implicitly encouraged by rate coding or focus on architectural heuristics that do not generalize to high-resolution event-based benchmarks, leaving spike activity and accuracy tightly coupled. Fixed spike-rate regularization does not fully resolve this issue: a coefficient chosen to suppress dense frame-encoded activity can over-constrain sparse event streams, while one chosen for native event inputs may not sufficiently penalize frame encodings as the replay distribution evolves. This limitation becomes acute when spike budgets must adapt to diverse input modalities—particularly when contrasting dense Poisson-encoded frames against naturally sparse event streams.

To bridge this gap, we propose an energy-aware spike budgeting framework for continual learning in SNNs. We evaluate our method across both frame-based (MNIST, CIFAR-10) and event-based (N-MNIST, CIFAR-10-DVS, DVS-Gesture) datasets. Our approach combines experience replay, learnable neuron dynamics, and an adaptive spike-budget controller to enforce dataset-specific spike-activity constraints. Rather than using a static sparsity weight, the controller closes a feedback loop by comparing observed mini-batch activity with a target budget during replay, allowing the same control law to tighten or relax the activity constraint as the task stream changes. On frame-based datasets, our method acts as a sparsity-inducing regularizer, reducing spikes by 47% on MNIST while improving accuracy by 2.31% points. Conversely, on event-based datasets like DVS-Gesture, relaxing the budget allows for a significant accuracy gain (17.45% points) with only a marginal increase in spike activity, demonstrating the importance of modality-adaptive spike-activity regulation. Our contributions are summarized as follows:

- **Energy-aware continual learning framework:** We introduce an adaptive spike budgeting mechanism based on proportional control that dynamically regulates network activity during continual learning, treating spike rate as an explicit activity-dependent proxy objective alongside accuracy rather than as a static sparsity penalty.
- **Modality-dependent duality discovery:** We reveal and characterize an operational duality in SNN continual learning: spike budgeting acts as sparsity regularization on frame-based datasets (preventing overfitting) while enabling controlled activation increase on event-based datasets (preventing underfitting).
- **Cross-modality evaluation and analysis:** We provide a comprehensive study of continual learning spanning five benchmarks across both frame-based and event-based vision, demonstrating consistent improvements in the accuracy–spike-activity trade-off and strong absolute performance (91.93%) on DVS-Gesture.

Our results highlight the importance of activity-aware, modality-adaptive learning, bringing continual SNN learning closer to practical neuromorphic deployment.

2. Preliminaries

This section provides the essential background required to understand our method. We briefly review the leaky integrate-and-fire (LIF) neuron model with surrogate-gradient learning, define the spike-rate activity proxy used throughout the paper, and summarize the continual learning evaluation metrics used in our experiments.

2.1. LIF neuron model and surrogate gradients

All networks in this work utilize discrete-time LIF neurons, simulated via the `snnTorch` framework [20]. The membrane potential $U[t]$ of a neuron at time step t evolves according to the following difference equation:

$$U[t+1] = \beta U[t] + I_{\text{in}}[t+1] - S[t] V_{\text{thr}}, \quad (1)$$

where $\beta \in (0, 1)$ is the membrane potential decay rate, $I_{\text{in}}[t + 1]$ is the synaptic input current (weighted sum of incoming spikes), and V_{thr} is the firing threshold. A spike $S[t] \in \{0, 1\}$ is emitted if the membrane potential exceeds the threshold:

$$S[t] = \Theta(U[t] - V_{\text{thr}}), \quad (2)$$

where $\Theta(\cdot)$ is the Heaviside step function. The term $-S[t]V_{\text{thr}}$ in equation (1) implements a soft reset (reset-by-subtraction) mechanism, which is known to preserve residual potential and mitigate information loss compared to hard resets [21]. To enable training via backpropagation through time (BPTT), we address the non-differentiability of the spike function using the Fast Sigmoid surrogate gradient [22], defined as:

$$\frac{\partial S}{\partial U} \approx \frac{1}{(1 + k|U - V_{\text{thr}}|)^2}, \quad (3)$$

where k is a slope parameter governing the steepness of the surrogate derivative. We use $k = 25$ (the `snnTorch` default) and favor the Fast Sigmoid because it offers a good balance between gradient stability and task accuracy [22]. This formulation ensures smooth gradient flow during the backward pass while maintaining binary spiking dynamics in the forward pass.

2.2. Spike-rate as an activity-dependent proxy

Consistent with neuromorphic hardware literature on Intel Loihi [4], IBM TrueNorth [23], and broader neuromorphic computing analyses [2], we use spike rate as an algorithmic indicator of activity-dependent synaptic-event cost. For each sample, the spike rate is computed as the average number of spikes per neuron per timestep:

$$\text{SpikeRate} = \frac{1}{NT} \sum_{n=1}^N \sum_{t=1}^T z_{n,t}, \quad (4)$$

where N is the number of neurons and T is the number of simulation timesteps. Throughout this paper, tabulated spike rates are reported in percent, i.e. $\text{SpikeRate}(\%) = 100 \times \text{SpikeRate}$; controller trace figures use the raw normalized rate so that activity can be compared directly against the target budget. This choice follows the hardware observation that synaptic operations triggered by active spikes dominate many activity-dependent costs in neuromorphic systems [4]. The metric focuses on the activity-dependent component relevant to algorithm design and does not include static leakage power. Frame-based datasets naturally induce denser spike activity through Poisson rate coding, while event-based datasets produce inherently ultra-sparse activations.

Spike rate is not a direct hardware power measurement: chip-level energy also depends on static leakage, routing, memory access, and peripheral circuitry. It should therefore be interpreted as the synaptic-event component that algorithm design can influence.

2.3. Continual learning metrics

We adopt standard continual-learning metrics following [24]. After training all tasks sequentially, the final average accuracy (ACC) is defined as:

$$\text{ACC} = \frac{1}{K} \sum_{k=1}^K A_{K,k}, \quad (5)$$

where $A_{K,k}$ denotes accuracy on task k after completing all K tasks. Forgetting (F) is measured as the reduction from each task's maximal observed accuracy:

$$F = \frac{1}{K-1} \sum_{k=1}^{K-1} \left(\max_{j \leq K} A_{j,k} - A_{K,k} \right). \quad (6)$$

Backward transfer (BWT) measures the influence of learning new tasks on the performance of previously learned tasks. It is defined as:

$$\text{BWT} = \frac{1}{K-1} \sum_{k=1}^{K-1} (A_{K,k} - A_{k,k}), \quad (7)$$

where $A_{k,k}$ denotes accuracy on task k immediately after it was learned. A positive BWT indicates that learning new tasks has improved the performance on older tasks, while a negative BWT implies catastrophic forgetting.

All models are implemented in PyTorch using `snnTorch` and trained end-to-end with surrogate-gradient backpropagation. Full architectural and training details are provided in appendix B.

3. Method: energy-aware continual learning in SNNs

3.1. Spiking network architectures

In this work, we employ a family of SNNs across all datasets, ensuring that every model uses LIF neurons trained with surrogate-gradient backpropagation. No ANN or non-spiking baselines are used; all architectures are fully spiking networks (with either convolutional or fully-connected backbones depending on the dataset).

Our architectures follow a unified design principle adapted to dataset complexity and input resolution. For low-resolution tasks (MNIST, Fashion-MNIST), we use a lightweight fully-connected SNN (Linear–LIF–Linear). For CIFAR-10, we use a VGG-style spiking CNN. For high-resolution event streams (CIFAR-10-DVS, DVS-Gesture), we use a deeper spiking CNN with a compact classifier head to capture fine-grained spatiotemporal features. Detailed layer specifications and tensor shapes are provided in appendix B.

3.2. Energy-aware spike budgeting framework

A core contribution of this work is the integration of an spike-budget controller that dynamically regulates network activity during training. Motivated by the activity constraints of neuromorphic hardware, where spike-triggered synaptic events are a major contributor to activity-dependent cost [4, 23], we introduce a mechanism to enforce a dataset-specific spike budget.

We formulate spike budgeting as a feedback control problem. This mechanism mimics homeostatic plasticity principles [25] by dynamically adjusting a regularization coefficient λ_{rate} based on the error between the current mini-batch spike rate r_{spike} and a target budget r_{target} :

$$\Delta\lambda_{\text{rate}} = \eta \cdot (r_{\text{spike}} - r_{\text{target}}), \quad (8)$$

where η is the controller gain. The regularization strength λ_{rate} is updated periodically and clipped to a predefined range $[\lambda_{\text{min}}, \lambda_{\text{max}}]$. This coefficient weights a penalty term in the loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda_{\text{rate}} \cdot (r_{\text{spike}} - r_{\text{target}})^2. \quad (9)$$

The squared penalty provides a bidirectional gradient with respect to r_{spike} : when $r_{\text{spike}} > r_{\text{target}}$, $\frac{\partial \mathcal{L}}{\partial r_{\text{spike}}} = 2\lambda_{\text{rate}}(r_{\text{spike}} - r_{\text{target}}) > 0$ suppresses activity; when $r_{\text{spike}} < r_{\text{target}}$, the gradient is negative and would favor increased activity. Crucially, the proportional controller modulates how strongly this term competes with the task loss: if spikes exceed the target, λ_{rate} increases, enforcing sparsity; if spikes fall short, λ_{rate} decreases, effectively relaxing the spike-activity constraint so that $\mathcal{L}_{\text{task}}$ can drive the amount of activity needed for accurate temporal feature extraction.

This mechanism exhibits modality-adaptive behavior. On dense frame-based datasets, where spike rates naturally exceed the target, the controller enforces sparsity. On ultra-sparse event-based datasets, where spike rates fall below the target, the bidirectional penalty gradient encourages increased activity to improve temporal feature extraction. We formalize this modality-dependent behavior as two operating regimes of the same proportional control law: a sparsity-suppression regime when $r_{\text{spike}} > r_{\text{target}}$ and a controlled-activation regime when $r_{\text{spike}} < r_{\text{target}}$. For detailed spike-rate measurement (appendix A.1), controller dynamics (appendix A.2), and practical tuning ranges (appendix A.4), see appendix A.

Difference from fixed spike-rate regularization. Conventional methods apply a static sparsity penalty that remains constant throughout training. In contrast, our controller closes the loop by measuring actual spike activity and adjusting λ_{rate} proportionally to the tracking error. This feedback mechanism is essential during continual learning because the activity distribution shifts as new tasks arrive: a fixed penalty optimized for early tasks may over- or under-constrain later ones, whereas the proportional update adapts without per-task tuning. The controller operates only at training time; at inference, the trained network runs with fixed weights and neuron parameters, incurring no runtime overhead.

3.3. Learnable neuron dynamics

In addition to the spike budgeting framework, we augment the network’s representational capacity through learnable LIF neuron parameters. Standard LIF neurons typically use fixed decay rates and thresholds. To enhance the network’s ability to capture diverse temporal dynamics—ranging from static frames to fast-changing event streams—we make the membrane decay parameter β and the firing threshold V_{thr} learnable. These are implemented as layer-wise scalars, adding only negligible parameter overhead ($O(L)$ rather than $O(N)$), ensuring the memory footprint remains virtually unchanged. This allows the network to adapt its temporal integration window and sensitivity to the specific statistics of the input data [26].

3.4. Continual learning components

Having established the spike-activity budgeting mechanism and learnable neuron dynamics, we now describe the continual learning protocol and memory consolidation strategy that complete our framework.

3.4.1. Class-incremental protocol

We adopt a standard class-incremental continual learning (Class-IL) protocol. The dataset is partitioned into a sequence of K disjoint tasks, $\mathcal{T} = \{T_1, T_2, \dots, T_K\}$, where each task contains a unique subset of classes. The model is trained sequentially on each task without access to task identifiers during inference. This setting is particularly challenging as it requires the model to discriminate between all classes seen so far without catastrophic forgetting.

3.4.2. Experience replay

To mitigate forgetting, we employ experience replay, a widely effective strategy in continual learning [11]. We maintain a fixed-size episodic memory buffer $\mathcal{M}$ that stores a class-balanced subset of samples from previous tasks. During training on task T_k , minibatches are constructed by concatenating current-task samples with a random subset drawn from $\mathcal{M}$. This ensures that the optimization objective approximates the joint distribution of all seen tasks, stabilizing the plasticity-stability trade-off. Replay budgets used across datasets are summarized in appendix B.4 (table B5).

3.5. Ablation framework

To systematically evaluate the contributions of each component, we define a hierarchy of experimental configurations that enable controlled ablation studies:

- **C0 (Naive baseline):** The SNN is trained sequentially on tasks without any replay or regularization. This serves as a lower bound to quantify the severity of catastrophic forgetting.
- **C1 (Replay baseline):** Adds the experience replay buffer to the naive baseline. This represents the standard approach for mitigating forgetting in SNNs.
- **C2 (Learnable dynamics):** Augments C1 with learnable LIF parameters (β, V_{thr}), isolating the benefit of adaptive temporal integration.
- **C3 (Spike-budget controller):** Augments C1 with the spike-budget controller, isolating the impact of spike budgeting on accuracy and spike activity.
- **C4 (Combined framework):** Integrates all components: experience replay, learnable dynamics, and the spike-budget controller. This represents our proposed holistic approach.

4. Experiments

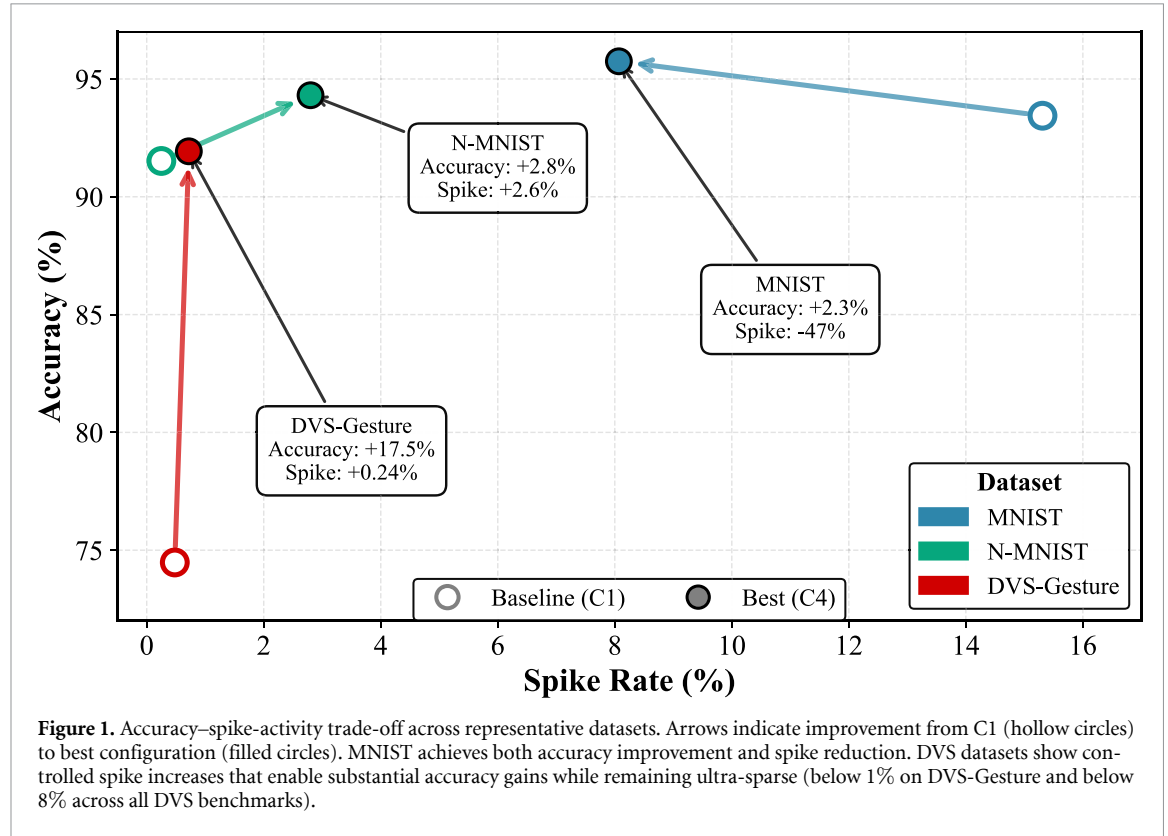
4.1. Experimental setup

We evaluate the proposed framework on five benchmarks spanning frame-based and event-based modalities: MNIST [27], N-MNIST [28], CIFAR-10 [29], CIFAR-10-DVS [30], and DVS-Gesture [31]. All experiments follow the class-incremental (Class-IL) protocol, a challenging continual learning setting where models encounter disjoint class subsets sequentially without access to task identifiers at inference [8, 24]. Table 1 summarizes the dataset configurations.

Tasks denotes the class-incremental schedule (e.g. 5×2 indicates five tasks with two classes each, whereas $4 + 4 + 3$ covers three tasks with 4, 4, and 3 classes). These timestep budgets reflect the typical temporal extent of each dataset: CIFAR-10-DVS clips comprise short fixed-duration event windows,

Table 1. Dataset specifications. T denotes timesteps for temporal integration.

Dataset	Res.	Classes	Tasks	T	Modality
MNIST	28×28	10	5×2	25	Frame (Poisson)
N-MNIST	34×34	10	5×2	50	DVS (events)
CIFAR-10	32×32	10	5×2	50	Frame (Poisson)
CIFAR-10-DVS	128×128	10	5×2	50	DVS (events)
DVS-Gesture	128×128	11	$4+4+3$	60	DVS (events)



whereas DVS-Gesture requires $T = 60$ to cover full gesture sequences without incurring excessive spike counts.

Implementation. Frame-based datasets are converted to spike trains via Poisson rate encoding [32], while event-based datasets retain their native DVS representation [5, 33]. All models employ LIF-based spiking architectures trained end-to-end with surrogate gradient backpropagation [34, 35]. The adaptive spike-budget controller enforces dataset-specific spike-activity constraints, providing hardware-relevant activity signals because synaptic operations are a major activity-dependent cost in neuromorphic processors [4, 23]. Replay buffer allocations are summarized in table B5 (appendix B.4): MNIST and N-MNIST use 2000 final stored samples (200 slots per class); CIFAR-10 and CIFAR-10-DVS use 10 000 samples (1000 slots per class); DVS-Gesture uses 440 samples (40 slots per class). During training on each new task, the current-task cross-entropy is augmented with a uniformly sampled replay cross-entropy term weighted by 0.5; replay minibatches are capped at 32 examples. No task identifier is provided at inference. All C1–C4 comparisons within each dataset use identical architectures and replay budgets, so the effects of spike budgeting and learnable dynamics are isolated from architectural or memory differences. Full architectural specifications plus hyperparameters are detailed in appendix B.

Evaluation metrics. Following standard continual learning methodology [12, 24], we report: (i) average accuracy across all tasks, (ii) forgetting, measured as mean accuracy degradation from peak performance, (iii) backward transfer (BWT), and (iv) spike rate as an activity-dependent synaptic-event proxy. Unless stated otherwise, spike rate (%) is computed during training as the mini-batch average activity (appendix A.1) and then averaged across the training process. All metrics are averaged over multiple random seeds (appendix C). Inference-time spike rates were consistent with training-time averages.

Table 2. Comparison of replay baseline (C1) with best configuration (C4). Bold indicates best performance.

Dataset	Cfg	Acc (%) ^a	Forgetting (%)	BWT (%)	Spike (%)
MNIST	C1	93.44 ± 1.40	7.17	−2.45	15.31
	C4	95.75 ± 0.16	4.40	−1.65	8.07
N-MNIST	C1	92.27 ± 2.43	6.09	−8.59	0.25
	C4	94.07 ± 1.35	4.06	−5.37	2.70
CIFAR-10	C1	59.50 ± 1.58	33.56	−60.37	37.50
	C4	61.26 ± 0.79	33.56	−60.16	30.90
CIFAR-10-DVS	C1	45.52 ± 4.24	37.11	−38.19	1.18
	C4	49.68 ± 0.93	29.51	−25.07	7.25
DVS-Gesture	C1	74.48 ± 0.74	25.35	−25.35	0.48
	C4	91.93 ± 1.85	5.90	−5.90	0.72

^a Mean ± sample standard deviation over the seed subsets detailed in appendix C.

4.2. Main results

Table 2 compares our best configurations against the experience replay baseline (C1) [11]. Without continual learning (C0, defined in section 3.5), catastrophic forgetting reduces accuracy to ~15%–19% across all datasets (see appendix C for per-seed evidence). Experience replay (C1) establishes a strong baseline, which our proposed framework further improves.

Results are averaged over multiple independent training trials to ensure statistical robustness. The substantial BWT improvements (−25.35% → −5.90% for DVS-Gesture) demonstrate the stability of our method across repeated runs.

Before unpacking the per-modality trends, figure 1 summarizes how each optimized configuration shifts relative to its replay baseline in the accuracy–spike-activity plane, making the emerging trade-offs explicit.

Modality-dependent behavior. Two distinct patterns emerge across modalities. On frame-based datasets (MNIST, CIFAR-10), spike budgeting acts as implicit regularization, improving accuracy while reducing the spike-rate proxy for activity-dependent synaptic cost. This sparsity-inducing effect is particularly pronounced on MNIST, where the 47% spike reduction occurs alongside accuracy improvements, suggesting that dense Poisson encoding induces redundant activations that can be pruned without information loss. On event-based datasets (N-MNIST, CIFAR-10-DVS, DVS-Gesture), controlled increases in spike activity yield substantial accuracy gains while maintaining ultra-sparse operation. This aligns with SNN theory [2, 36]: event-driven inputs require sufficient network activation for temporal feature extraction, and overly aggressive sparsity constraints undermine the temporal dynamics necessary for processing asynchronous event streams.

Cross-dataset magnitude analysis. The magnitude of improvement varies significantly across datasets. DVS-Gesture exhibits the largest accuracy gain (17.45% points), several times larger than gains on other datasets. This exceptional improvement stems from three factors: (1) high temporal complexity of gesture sequences requiring adaptive dynamics, (2) relatively low replay baseline (74.48%) leaving substantial room for improvement, and (3) optimal synergy between learnable LIF parameters and spike budgeting for moderate temporal sequences ($T = 60$). In contrast, N-MNIST shows modest gains (1.80% points) from a higher baseline (92.27%), with forgetting reduced from 6.09% to 4.06%. This suggests that simpler event-based classification tasks approach performance saturation with standard replay mechanisms, though still benefit from reduced forgetting.

Configuration selection. Across modalities, C4 provides the best accuracy–spike-activity compromise by combining replay, learnable dynamics, and spike budgeting [26, 37]. CIFAR-10 still benefits primarily from the controller term, while the learnable neuron parameters act as a neutral regularizer, illustrating that the unified configuration remains effective even when individual components contribute unevenly. This behavior reinforces the broader design principle: adaptive temporal parameters are most valuable for highly dynamic inputs (DVS streams) but remain compatible with frame-based data when spike budgeting curbs any excess capacity.

Table 3. Ablation study on MNIST (5 tasks $\times$ 2 classes).

Configuration	Acc. (%) ^a	Forgetting (%)	Spike (%)
C0 (Naive)	19.35 $\pm$ 0.48	99.76	15.31
C1 (Replay)	93.44 $\pm$ 1.40	7.17	15.31
C2 (Replay + learnable)	94.25 $\pm$ 0.83	6.16	15.31
C3 (Replay + controller)	92.40 $\pm$ 1.75	8.39	10.47
C4 (All components)	95.75 $\pm$ 0.16	4.40	8.07

^a Mean $\pm$ sample standard deviation over three seeds (appendix C).**Table 4.** Diagnostic component ablation on N-MNIST (5 tasks $\times$ 2 classes, $T = 25$).

Configuration	Acc. (%) ^a	Forgetting (%)	BWT (%)	Spike (%)
C0 (Naive)	15.82	91.17	−61.86	1.00
C1 (Replay)	93.80	6.07	−0.13	1.00
C2 (Replay + learnable)	93.69	6.25	−0.11	1.00
C3 (Replay + controller)	93.96	5.91	−0.16	7.97
C4 (All components)	94.76	4.91	0.11	6.56

^a Mean over three independent runs per configuration.

4.3. Ablation study

To quantify the contribution of each framework component, we conduct systematic ablation studies. Table 3 presents results on MNIST, illustrating the progression from naive sequential training to our complete method.

Experience replay [11] is essential, eliminating catastrophic forgetting [6] and establishing a strong 93.44% baseline. The dramatic improvement from C0 to C1 (74.09% points) confirms that replay-based memory consolidation is non-negotiable for continual SNN learning. Learnable LIF parameters (C2) provide a modest accuracy gain (+0.81% points) without increasing spike activity, suggesting that adaptive temporal integration enhances feature discrimination without inducing additional spiking. This no-added-spike property makes learnable dynamics an appealing augmentation, as it improves plasticity without compromising the sparsity advantages of SNNs.

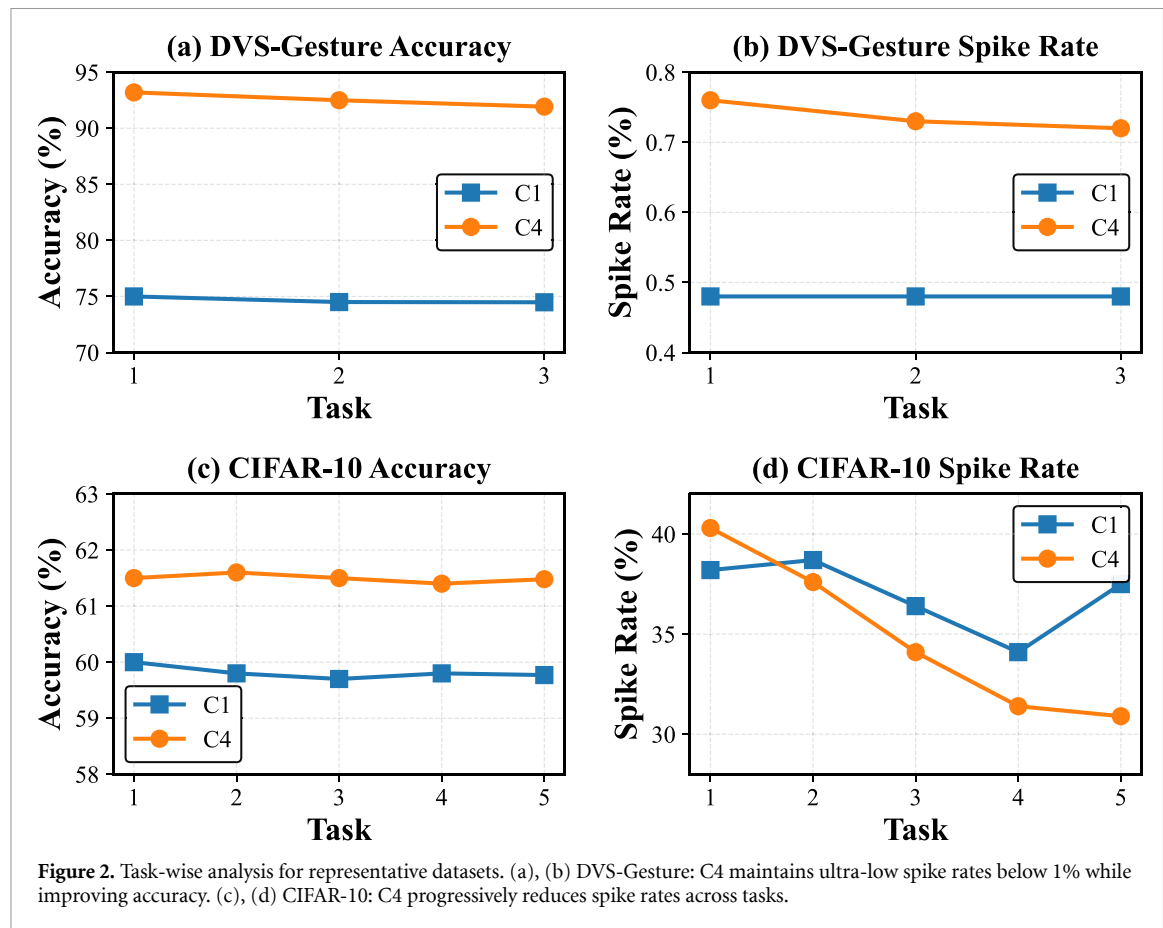
The controller alone (C3) reduces spikes by 32% but marginally decreases accuracy (−1.04% points), revealing an accuracy–spike–sparsity trade-off when applied in isolation. However, combining all components (C4) achieves synergistic improvements: highest accuracy (+2.31% points over C1), lowest forgetting (−2.77% points over C1), and 47% spike reduction. This synergy suggests that learnable dynamics partially compensate for the reduced expressiveness imposed by spike constraints, while the controller’s regularization effect mitigates potential overfitting from increased parameter flexibility. The non-additive nature of these improvements underscores the importance of holistic co-design in neuromorphic systems. Full ablation results across all datasets are in appendix C.

To validate whether the component interactions observed on frame-based data generalize to the event-data regime, we run a diagnostic full C0–C4 ablation on N-MNIST, an event-based DVS dataset, using the same five-task split and replay capacity but a shorter $T = 25$, five-epoch schedule for the full component sweep. The $T = 50$ C1/C4 results in table 2 remain the primary N-MNIST benchmark; table 4 isolates component interactions under this tractable diagnostic setting.

The results mirror the MNIST pattern. Replay alone (C1) eliminates catastrophic forgetting, raising accuracy from 15.82% to 93.80%. Learnable LIF parameters (C2) maintain comparable accuracy (93.69%) without increasing spike activity, while the controller (C3) adds a modest accuracy improvement (93.96%) by permitting controlled activity increases. C4 achieves the best overall accuracy (94.76%) and lowest forgetting (4.91%), confirming the same non-additive synergy. Notably, the controller in C3 and C4 raises spike activity from the native event-stream floor (1.00%) to a controlled range (6.56%–7.97%)—still below a 10% activity envelope—demonstrating that the same proportional control law adapts appropriately to event-based input statistics without modality-specific re-engineering.

4.4. Analysis by modality

Event-based datasets. DVS sensors produce inherently sparse, asynchronous event streams [33]. On DVS-Gesture [31], C4 achieves dramatic accuracy gains with minimal spike increase (figures 2(a) and (b)). Task-wise analysis reveals that C4 maintains consistent improvements across all gesture groups,

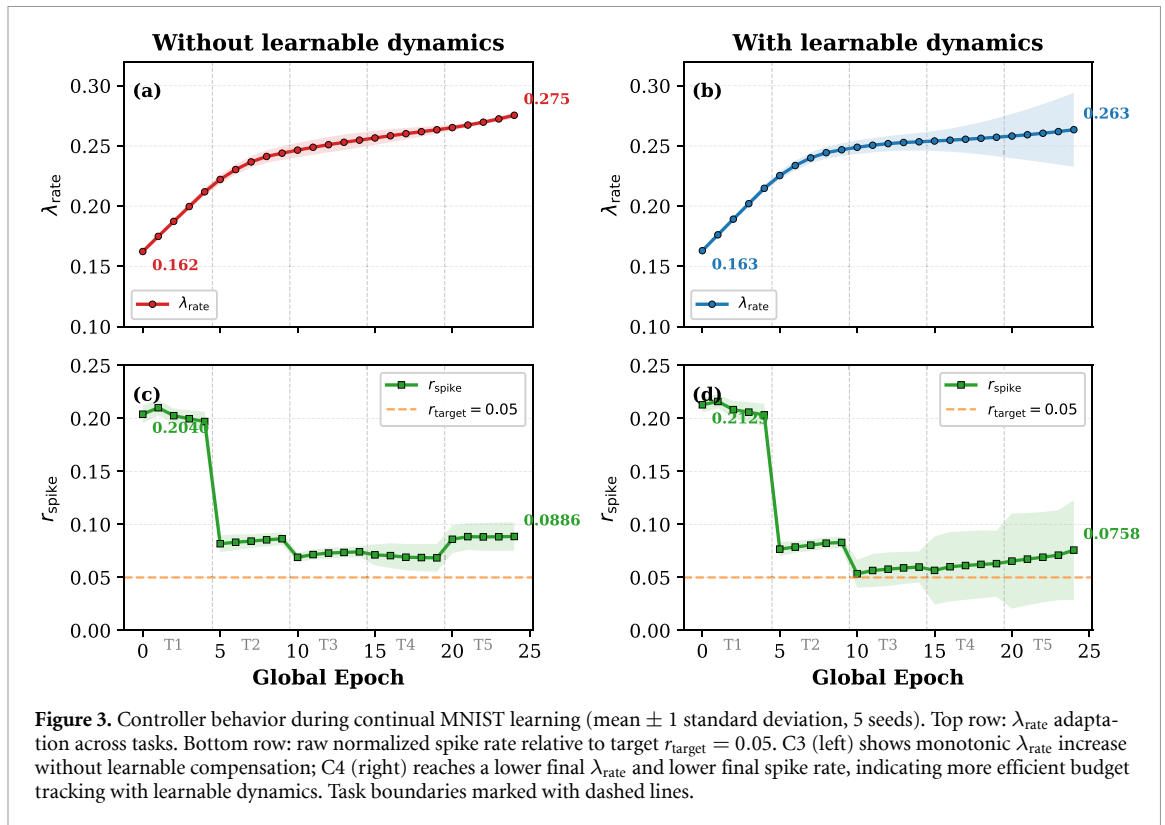


with particularly strong performance on the final task (Task 3), which exhibits the most challenging 3-class discrimination problem. The spike rate remains below 1% throughout sequential learning, demonstrating that the controlled activation increase ($0.48\% \rightarrow 0.72\%$) operates well within the ultra-sparse regime characteristic of event-driven neuromorphic operation [2].

Notably, the absolute spike increase ($+0.24\%$ points) is minimal compared to frame-based spike reductions (-7.24% points on MNIST), indicating that the controller adapts its behavior to input statistics rather than enforcing uniform sparsity constraints. This adaptive property is critical for DVS datasets, where excessive sparsity enforcement can eliminate the temporal information encoded in event timing. The bidirectional gradient mechanism naturally discovers this balance: when spike rates fall below the target, the penalty gradient weakens, allowing the classification loss to drive learning while maintaining hardware-compatible operation.

Frame-based datasets. Poisson-encoded frame data naturally induces denser spike activity. On CIFAR-10, C4 yields a 1.76%-point accuracy improvement and 17.6% spike reduction reported in table 2, with the controller component supplying most of the gain and the learnable parameters acting as a neutral regularizer (figures 2(c) and (d)). Task-wise analysis shows that C4 achieves progressive spike reduction across sequential tasks, with particularly strong compression on later tasks (Tasks 4–5), where the replay buffer provides richer supervision. This progressive improvement suggests that the controller learns effective regularization as the task distribution stabilizes with experience.

The 17.6% spike reduction on CIFAR-10 is less pronounced than MNIST’s 47% reduction, reflecting the inherently higher complexity of 32×32 RGB natural images compared to 28×28 grayscale digits. Despite this, the spike rate remains substantially lower than typical dense ANNs would induce via rate encoding, confirming that spike budgeting successfully exploits redundancy in Poisson-encoded representations. Importantly, accuracy improves alongside sparsity, contradicting the intuitive expectation of an accuracy–sparsity trade-off and validating our hypothesis that redundant spikes can impair generalization through effective overfitting.



4.5. Controller behavior and stability

To characterize the controller's dynamics and justify the use of a purely proportional control law, we analyze the evolution of λ_{rate} and the spike rate r_{spike} during MNIST continual training, averaged over 5 seeds (figure 3).

Three properties emerge from the traces. **First**, the controller operates stably within its clipping bounds $[0, 0.8]$, never hitting either limit. Updates are driven by batch-normalized spike-rate error with gain $\eta = 0.08$, producing smooth, oscillation-free adaptation across all 25 epochs.

Second, the coupling with learnable dynamics is essential for favorable controller behavior. Without learnable LIF parameters (C3), λ_{rate} increases monotonically from 0.162 to 0.275 because spike rates remain above the target and the model lacks capacity to compensate. With learnable β and V_{thr} (C4), the controller reaches a lower final λ_{rate} (0.263 vs 0.275) while achieving a lower final spike rate (0.076 vs 0.089) and a smaller mean tracking error (0.046 vs 0.053). The adaptive temporal parameters help the network satisfy the spike budget with lower tracking error, so the controller does not need to tighten as aggressively as in C3. This interaction explains the synergistic accuracy gain in the ablation: the controller enforces the budget, while learnable dynamics supply the degrees of freedom to meet both the budget and the task objective.

Third, we justify the choice of a purely proportional controller. An integral term would accumulate state across task boundaries, risking destabilization when the data distribution changes abruptly between tasks in continual learning. The proportional gain is sufficient for epoch-scale adaptation of slowly varying activity levels, and the clipping bounds provide a practical boundedness guarantee for λ_{rate} rather than a formal closed-loop stability proof. The controller is a training-time mechanism only: at inference, the trained network runs with fixed weights and neuron parameters, so no runtime overhead is incurred.

4.6. Accuracy–spike–activity trade-offs

Figure 1 visualizes the accuracy–spike–activity landscape across representative datasets, revealing fundamental differences in optimization trajectories. On MNIST, the improvement vector from C1 to C4 points toward the upper-left quadrant—simultaneous accuracy improvement and spike-rate reduction—a rare and desirable outcome in machine learning. This Pareto improvement demonstrates that spike budgeting on frame-based data does not merely trade accuracy for sparsity but actually enhances both objectives through implicit regularization.

DVS datasets exhibit a different trajectory: improvements vector toward the upper-right quadrant, trading small-to-moderate absolute spike-rate increases for substantial accuracy gains. However, the magnitude of these trade-offs varies dramatically. DVS-Gesture achieves a highly favorable 72:1 accuracy-to-spike ratio (17.45%-point accuracy gain / 0.24%-point spike increase), while CIFAR-10-DVS shows a less favorable 0.68:1 ratio (4.16%-point accuracy gain / 6.07%-point spike increase). This disparity reflects the temporal complexity of gesture recognition versus object classification: gesture sequences ($T = 60$) benefit more from increased representational capacity than shorter CIFAR-10-DVS event windows ($T = 50$).

Hardware implications. Neuromorphic processors such as Loihi [4] and TrueNorth [23] exhibit activity-dependent costs that are strongly influenced by spike-triggered synaptic operations. Our 47% relative spike reduction on MNIST (15.31% $\rightarrow$ 8.07%) therefore implies a commensurate reduction in activity-driven synaptic events, all else equal. On DVS-Gesture, despite the +0.24%-point spike increase, absolute spike rates remain below 1%, maintaining compatibility with sparse neuromorphic operating regimes where event rates $<10\%$ are considered efficient [2]. These results demonstrate that spike-activity-aware training during continual learning preserves algorithmic deployment viability on resource-constrained neuromorphic platforms. The reported spike-rate changes should be interpreted under a fixed algorithmic mapping rather than as direct chip-level energy or power measurements. The controller itself adds a scalar update and spike-counting step during training but imposes no inference-time overhead on the deployed SNN. Verifying actual chip-level energy or power effects requires deployment on specific neuromorphic processors, where hardware-dependent factors including process technology, dataflow mapping, and peripheral power become relevant—directions we leave to future work.

5. Discussion

The empirical study shows that energy-aware spike budgeting is not merely a regularizer but a control primitive that reshapes the plasticity–stability trade-off in modality-dependent ways. On frame-based datasets, the controller consistently drives C4 toward lower spike regimes than replay-only baselines while preserving or improving accuracy, indicating that redundant Poisson activations can be trimmed without hurting separability. Conversely, event-based datasets benefit from carefully relaxing the budget: even modest increases in spike rate (0.48% $\rightarrow$ 0.72% on DVS-Gesture) supply the temporal support required for long gestures while keeping absolute activity well inside neuromorphic sparsity envelopes. This dual behavior validates our claim that spike activity should be treated as a first-class algorithmic control signal rather than a passive byproduct of SNN sparsity.

To contextualize these results against existing methods, table 5 provides a compact protocol-aware comparison with recent representative spiking continual-learning studies. These methods are typically evaluated under different protocols, architectures, and datasets, so direct numerical ranking is not meaningful. We therefore use the comparison to position the proposed framework as complementary: it focuses on the accuracy–spike-activity trade-off in continual spiking vision across both frame-based and native event-based datasets, and shows how explicit spike-budget control can reshape forgetting behavior while keeping activity within neuromorphic sparsity envelopes.

Beyond this contextual comparison, the ablation analysis (table 3) clarifies why combining replay, learnable neuron dynamics, and the controller is essential. Replay alone removes catastrophic forgetting but leaves excess spike activity on frame data; the controller alone enforces budgets yet suffers accuracy loss; learnable dynamics introduce temporal flexibility but require an activity-aware constraint to remain stable. C4 leverages all three: replay supplies representative gradients, learnable β/V_{thr} tune per-dataset time constants, and the controller enforces spike-activity bounds. This decomposition suggests a design rule for future systems—pair any capacity-expanding mechanism (e.g. adaptive neurons or dendritic compartments) with an explicit activity controller to keep continual learning trajectories within feasible regions.

From a deployment standpoint, the learned controller resembles a software-defined activity constraint for neuromorphic hardware. Because the budget is enforced during training, inference-time activity remains predictable without extra calibration, easing mapping onto Loihi- or TrueNorth-like substrates where synaptic operations are a major component of activity-dependent cost [4, 23]. The ability to dial spike targets per modality further enables heterogeneous sensor suites: a single policy can throttle Poisson-coded cameras aggressively while granting additional headroom to asynchronous event sensors, simplifying firmware integration in mixed-vision platforms.

Table 5. Main-text protocol-aware context for recent spiking continual-learning studies. Reported metrics are taken from the cited papers and are not directly comparable across rows because benchmarks and continual-learning settings differ.

Work	Setting / benchmark	Reported metric(s)	Scope relative to this paper
HLOP [38]	Permuted-MNIST, domain-incremental	ACC 95.15%, BWT -1.30%	Strong retention on non-DVS domain-incremental tasks; no explicit spike-budget controller or event-vision energy analysis.
TACOS [39]	Split-MNIST / Split-FMNIST, task-agnostic	Mean accuracy 82.56% / 93.22%	Task-agnostic domain-incremental setting with different data and architecture; no native DVS benchmark evaluation.
SOR-SNN [40]	CIFAR-100 class-incremental variants	Example 10-step ACC 80.12%	Targets pathway reorganization and memory/energy trade-offs on large-scale frame data rather than cross-modality DVS vision.
CoLaNET [41]	Permuted-MNIST, 10 tasks	Average accuracy $\approx 92.31\%$	Biologically plausible local-learning SNN on permuted-MNIST; no spike-budget feedback control across frame and DVS modalities.
This work	MNIST, CIFAR-10, N-MNIST, CIFAR-10-DVS, DVS-Gesture; class-incremental	Full framework improves ACC on all five benchmarks; DVS-Gesture reaches 91.93% with 0.72% spikes	Focuses on explicit accuracy-spike-activity control for continual SNN vision across both frame-based and native event-based datasets.

Despite these advantages, several limitations remain. Replay buffers still add non-trivial memory overhead, which may be challenging for on-chip deployment, and our controller currently relies on hand-tuned gains (η , $\lambda_{\max}$) per modality. We focus on class-incremental protocols with balanced task boundaries; task-free or open-world settings may require adaptive targets that respond to concept drift. Additionally, different datasets require different architectures to accommodate varying input resolutions and modality types, so absolute cross-dataset comparisons of spike rates and accuracies are confounded by architectural depth, channel count, and timestep budget. Our primary claims rest on within-dataset comparisons (C0–C4), where the architecture is held fixed; cross-dataset analyses characterize modality-dependent trends rather than asserting architecture-independent absolute superiority. Addressing these issues—through automatic gain tuning, generative memory surrogates, adaptive spike targets informed by streaming statistics, or controlled same-architecture experiments across modalities—remains fertile ground for future neuromorphic continual learning research.

6. Conclusion

We presented an energy-aware continual learning framework for SNNs that combines replay, learnable LIF parameters, and a proportional spike-budget controller. Across five benchmarks spanning frame-based and event-based vision, this unified approach consistently improved accuracy while either reducing or strategically increasing spike activity, demonstrating that spike-activity budgets can serve as an explicit algorithmic control signal rather than an afterthought. The modality-dependent behavior-sparsity regularization for dense Poisson inputs versus controlled activation boosts for ultra-sparse DVS streams—highlights the necessity of treating activity constraints as data-aware design parameters.

Looking ahead, we plan to deploy the controller-trained models on neuromorphic hardware to measure chip-level energy and power under concrete hardware mappings, investigate automatic tuning of budget targets for task-free or open-world streams, and pair the controller with lightweight generative replay to reduce memory footprints. These steps will further close the gap between algorithmic continual learning advances and practical neuromorphic perception systems.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Appendix A. Spike-activity budgeting details

Configurations C3 and C4 share the same spike-budget controller but express different ablations of our framework. This appendix documents the measurement strategy, control law, and operating regimes that enable the controller to remain stable across modalities.

A.1. Spike-rate measurement

We monitor mini-batch activity via

$$r_{\text{spike}} = \frac{N_{\text{spk}}}{NTB}, \quad (\text{A.1})$$

where N_{spk} counts spikes, N is the number of monitored neurons, T is the integration window, and B is the batch size. We use raw normalized r_{spike} in controller equations and trace plots, while tabulated summaries report $100 \times r_{\text{spike}}$ as spike rate (%). This normalized activity measure matches the spike-rate proxy in section 2 and is computed once per optimizer step so that controller updates keep pace with weight updates.

A.2. Budget loss and controller

With target budget r_{target} , the C3/C4 loss is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda_{\text{rate}} (r_{\text{spike}} - r_{\text{target}})^2, \quad (\text{A.2})$$

with a clipped proportional controller

$$\lambda_{\text{rate}} \leftarrow \text{clip}(\lambda_{\text{rate}} + \eta(r_{\text{spike}} - r_{\text{target}}), [\lambda_{\text{min}}, \lambda_{\text{max}}]). \quad (\text{A.3})$$

Whenever the network overshoots the budget, λ_{rate} increases and the penalty term suppresses spikes; once activity drops below the target, the controller gives back capacity so that the cross-entropy term dominates. In practice, we update λ_{rate} after each optimizer step and cache the resulting value so that the next batch starts from the most recent state.

A.3. Dataset-specific behavior

Frame-based datasets. C1 produces 15.31% spikes on MNIST; the controller steers C4 to 8.07% spikes (47% reduction) while improving accuracy by 2.31% points. The same trend holds for CIFAR-10, where the controller acts primarily as a regularizer.

Event-based datasets. DVS-Gesture begins at 0.48% spikes. Allowing a controlled increase to 0.72% unlocks 17.45% points of accuracy while keeping absolute activity below 1%. CIFAR-10-DVS similarly trades a modest spike increase for 4.16% points of accuracy, albeit with a less favorable accuracy-to-spike ratio because of the shorter temporal windows ($T = 50$).

A.4. Practical settings

For the reported C3/C4 experiments, we use a normalized target $r_{\text{target}} = 0.05$ (5% activity), initial $\lambda_{\text{rate}} = 0.15$, proportional gain $\eta = 0.08$, and clipping bounds $\lambda_{\text{rate}} \in [0, 0.8]$. These values are held fixed across the reported controller ablations unless a diagnostic setting is explicitly noted. Lower targets can silence useful neurons, whereas higher values erode the spike-sparsity advantages. The controller's moving-average window of 5 batches strikes a balance between responsiveness and stability; shorter windows react too aggressively on CIFAR-10-DVS.

A.5. Key takeaways

- The same control law operates across modalities because the spike-rate measure normalizes by both hidden units and timesteps, enabling a shared r_{target} interpretation.
- Replay keeps the controller honest: batches always mix old and new samples, so the measured spike rate reflects the deployed distribution rather than a single-task snapshot.
- Stability depends primarily on the proportional gain η and the clipping ceiling λ_{max} . In the reported experiments, fixed values ($\eta = 0.08$, $\lambda_{\text{max}} = 0.8$) were sufficient for the frame and DVS settings considered here.

Appendix B. Implementation and training details

Reproducing the study requires four ingredients: how tasks are constructed, which architectures pair with each modality, which optimizer settings unlock stable training, and how the replay buffer is provisioned. The following subsections cover those items in that order so the experimental pipeline can be mirrored end-to-end.

B.1. Dataset and task construction

Each dataset follows the class-incremental (Class-IL) protocol described in section 3.4.1: five sequential two-class tasks for MNIST, CIFAR-10, and N-MNIST; five event-based tasks for CIFAR-10-DVS; and a three-stage (4/4/3 gesture) split for DVS-Gesture. Frame-based benchmarks use Poisson rate encoding with $T = 25$ (MNIST) or $T = 50$ (CIFAR-10). Fashion-MNIST follows the same protocol as MNIST and is included for supplementary validation in appendix C (table C6). Event-based datasets retain their native polarity streams, using $T = 50$ for N-MNIST and CIFAR-10-DVS and $T = 60$ for the longer DVS-Gesture sequences. The budgeting targets in appendix A are chosen to match these inherent time scales. Under this protocol, the network never revisits earlier tasks during training; by the time CIFAR-10 reaches the fifth task it must already distinguish all ten classes without task labels. The higher timesteps on DVS-Gesture (60 vs 50) stem from the longer gesture duration, which would otherwise be truncated and lead to artificial forgetting unrelated to model capacity.

B.2. Model configurations

The continual-learning variants C0–C4 combine replay, learnable neuron dynamics, and spike budgeting as summarized in table B1. The architectures themselves follow a consistent motif: lightweight fully-connected SNNs for 28×28 inputs, VGG-style networks for CIFAR-10, and deeper four-block ConvSNNs for the high-resolution DVS datasets. Tables B2–B4 list the layer layouts used throughout the paper.

Table B1. Component mapping for configurations C0–C4.

Config	Replay	Learnable LIF	Spike-budget controller
C0	✗	✗	✗
C1	✓	✗	✗
C2	✓	✓	✗
C3	✓	✗	✓
C4	✓	✓	✓

C0 therefore quantifies catastrophic forgetting, C1 injects replay, C2 adds learnable LIF parameters, C3 activates the spike-budget controller, and C4 inherits every mechanism. This explicit mapping becomes important when interpreting later sections: any delta between rows in table 2 can be traced back to toggling a single column here.

Table B2. Fully-connected spiking architecture for MNIST and Fashion-MNIST.

Layer	Parameters	Output Size	LIF
Linear	$784 \rightarrow 128$	128	—
LIF	β, V_{thr} learnable (C2/C4)	128	✓
Linear	$128 \rightarrow 10$	10	—

Listing the architectures per modality prevents ambiguity about channel counts, LIF placements, and flatten dimensions—all of which influence spike statistics. For instance, the DVS model’s flatten layer at 16 384 features is the main contributor to spike activity; the controller must therefore regulate activity at that interface, whereas CIFAR-10 relies mainly on convolutional sparsity.

B.3. Optimization and training regime

All models are trained with Adam. Frame-based datasets use a learning rate of 10^{-3} , batch size 64, and five epochs per task; CIFAR-10 uses the same rate but ten epochs with cosine annealing. Event-based datasets use batch size 16; N-MNIST trains for forty epochs, CIFAR-10-DVS for thirty, and DVS-Gesture

Table B3. VGG-5 Spiking CNN architecture for CIFAR-10. BN denotes batch normalization.

Layer	Parameters	Output size	LIF
Conv2d + BN	$3 \rightarrow 64, k = 3, \text{pad} = 1$	32×32	—
Conv2d + BN	$64 \rightarrow 64, k = 3, \text{pad} = 1$	32×32	—
MaxPool2d	$k = 2$	16×16	—
LIF	β, V_{thr}	16×16	✓
Conv2d + BN	$64 \rightarrow 128, k = 3, \text{pad} = 1$	16×16	—
Conv2d + BN	$128 \rightarrow 128, k = 3, \text{pad} = 1$	16×16	—
MaxPool2d	$k = 2$	8×8	—
LIF	β, V_{thr}	8×8	✓
Conv2d + BN	$128 \rightarrow 256, k = 3, \text{pad} = 1$	8×8	—
MaxPool2d	$k = 2$	4×4	—
LIF	β, V_{thr}	4×4	✓
Flatten	—	4096	—
Linear	$4096 \rightarrow 10$	10	—

Table B4. Spiking CNN architecture for CIFAR-10-DVS and DVS-Gesture.

Layer	Parameters	Output Size	LIF?
Conv2d	$2 \rightarrow 32, k = 3, \text{pad} = 1$	128×128	—
MaxPool2d	$k = 2$	64×64	—
LIF	β, V_{thr}	64×64	✓
Conv2d	$32 \rightarrow 64, k = 3, \text{pad} = 1$	64×64	—
MaxPool2d	$k = 2$	32×32	—
LIF	β, V_{thr}	32×32	✓
Conv2d	$64 \rightarrow 128, k = 3, \text{pad} = 1$	32×32	—
MaxPool2d	$k = 2$	16×16	—
LIF	β, V_{thr}	16×16	✓
Conv2d	$128 \rightarrow 256, k = 3, \text{pad} = 1$	16×16	—
MaxPool2d	$k = 2$	8×8	—
LIF	β, V_{thr}	8×8	✓
Flatten	—	16 384	—
Linear	$16\,384 \rightarrow C$	C	—

for sixty epochs with 3×10^{-3} learning rate and a Reduce-on-Plateau scheduler. Gradient clipping (max norm = 1.0) stabilizes the deeper ConvSNNs, and each configuration completes within a few GPU-hours on a single modern accelerator, including replay sampling and controller updates. Longer training schedules were evaluated for DVS-Gesture but provided diminishing returns once the controller converged; rather than add a fourth control knob, we fix these hyperparameters for all seeds so that variability stems solely from data order and stochastic updates.

B.4. Replay buffer budgets

The episodic memory $\mathcal{M}$ that powers experience replay (section 3.4) is allocated per dataset before training and then partitioned uniformly across classes by the replay buffer. Table B5 enumerates the capacities used for all reported results; the per-class slot count equals $\lfloor |\mathcal{M}|/C \rfloor$ for C classes. For reference, a buffer of 10 000 samples corresponds to 20% of the CIFAR-10 training set (50 000 images). Frame-based datasets need larger buffers to counter intra-class variety, whereas DVS-Gesture prioritizes balanced coverage of its 11 gestures while keeping memory footprint modest. A systematic sweep over smaller buffers is left to future work; in this paper we keep the replay budget fixed per dataset to isolate the effect of spike budgeting and learnable dynamics.

Table B5. Replay memory allocation per dataset.

Dataset	Classes	Buffer Size	Slots / Class
MNIST / Fashion-MNIST	10	2000	200
N-MNIST	10	2000	200
CIFAR-10	10	10 000	1000
CIFAR-10-DVS	10	10 000	1000
DVS-Gesture	11	440	40

Appendix C. Extended results and ablations

The quantitative evidence that feeds table 2 is unpacked here via seed-level tables. Each table lists accuracy, forgetting, spike rate, and—where available—BWT so readers can trace how the aggregated numbers were formed.

C.1. Seed-level tables

We list every seed considered per dataset. The tables surface both accuracy/spike trends and BWT statistics where logs provide per-task information, illustrating the variability inherent to deep SNN training and the variability encountered on event-based benchmarks.

C.1.1. DVS-Gesture

Table C1 reports per-seed DVS-Gesture metrics under the replay baseline (C1) and the full framework (C4).

Table C1. DVS-Gesture per-seed results.

Seed	Config	Final Acc. (%)	Forgetting (%)	Spike Rate (%)
42	C1	75.00	25.87	0.48
42	C4	93.23	5.94	0.76
44	C1	73.96	24.83	0.48
44	C4	90.62	5.86	0.67

DVS-Gesture therefore illustrates the regime where relaxing the spike budget is essential: the shift from the replay-only baseline to C4 yields roughly 17%–18% points of accuracy while absolute spike rates remain below 1%, so the additional spike activity remains small in the reported proxy.

C.1.2. CIFAR-10-DVS

Table C2 summarizes performance on CIFAR-10-DVS. The variability across seeds highlights the challenge of training on this dataset; however, C4 consistently reduces forgetting in successful runs.

Table C2. CIFAR-10-DVS per-seed results.

Seed	Final Acc. (%)		Forgetting (%)		Spike Rate (%)
	C1	C4	C1	C4	C4
42	42.53	49.03	42.95	29.83	6.51
44	46.36	45.57	34.82	31.92	6.82
45	48.52	50.34	31.27	29.19	8.00

Across CIFAR-10-DVS seeds, C4 narrows forgetting by roughly 3%–13% points whenever training remains stable. This table provides the granular evidence behind the average BWT improvement reported in the main text.

C.1.3. N-MNIST

Table C3 shows final accuracy and spike rate on N-MNIST for both C1 and C4.

Table C3. N-MNIST per-seed results for C1 and C4.

Seed	Config	Final Acc. (%)	Spike Rate (%)
42	C1	93.59	0.25
42	C4	95.60	2.74
44	C1	89.46	0.25
44	C4	93.03	2.85
46	C1	93.76	0.25
46	C4	93.58	2.51

N-MNIST remains largely spike-limited: all C1 seeds operate at 0.25% spikes, whereas C4 permits a small activity increase that converts to 2%–4% points of accuracy on the more stable runs. The variability shown here explains the larger standard deviations listed in table 2.

C.1.4. MNIST

Table C4 details the results for MNIST across all configurations.

Table C4. MNIST per-seed results for C0–C4.

Seed	Config	Final Acc. (%)	Forgetting (%)	Spike Rate (%)
42	C0	19.84	99.82	15.31
43	C0	19.33	99.60	15.31
44	C0	18.89	99.86	15.31
42	C1	94.93	5.54	15.31
43	C1	93.24	7.14	15.31
44	C1	92.16	8.82	15.31
42	C2	92.22	8.71	15.31
43	C2	95.77	4.11	15.31
44	C2	94.76	5.65	15.31
42	C3	93.07	7.87	10.87
43	C3	90.41	10.69	10.31
44	C3	93.71	6.62	10.24
42	C4	95.59	4.70	7.64
43	C4	95.75	4.23	7.98
45	C4	95.91	4.27	8.59

The MNIST table exposes how each component contributes per seed: replay (C1) already reaches > 93% accuracy, learnable dynamics (C2) maintain the same spike rate while slightly improving fidelity, and the controller (C3) alone enforces sparsity. Combining them (C4) yields the consistent 95.7%–95.9% plateau highlighted in the main paper.

C.1.5. CIFAR-10 (Frame-Based)

Table C5 shows seed-level results for CIFAR-10 under the full C4 configuration. The controller remains the dominant contributor on this benchmark, so the aggregated statistics match the improvements highlighted in the main table.

Table C5. CIFAR-10 per-seed results for C1 and C4.

Seed	Final Acc. (%)		Forgetting (%)		Spike Rate (%)
	C1	C4	C1	C4	C4
42	60.95	60.35	32.49	32.49	30.90
43	57.81	61.72	33.15	33.15	30.90
44	59.74	61.72	35.05	35.05	30.90

On CIFAR-10, the controller reduces average spike rate (table 2) while maintaining or improving accuracy relative to replay-only training. In these runs, forgetting remains comparable between C1 and C4, suggesting that spike-activity control mainly improves generalization and sparsity without destabilizing replay-based consolidation.

C.1.6. Fashion-MNIST (supplementary validation)

Fashion-MNIST serves as an additional frame-based benchmark to validate the spike-sparsity findings observed on MNIST. Table C6 details the results. While accuracy improvements are modest compared to DVS datasets, the spike-rate reduction (up to 63% in C4) is substantial even on this supplementary split, reinforcing the sparsity-inducing regularization effect on frame-based vision tasks.

Fashion-MNIST behaves similarly to MNIST but with lower absolute accuracy. The key observation is that controller-heavy settings (C3/C4) cut spike activity by up to 63% while holding accuracy within ± 2 percentage points of the replay baseline, confirming that the proposed controller generalizes to supplementary datasets.

Table C6. Fashion-MNIST per-seed results for C0–C4.

Seed	Config	Final Acc. (%)	Forgetting (%)	Spike Rate (%)
42	C0	19.87	98.95	32.74
43	C0	19.88	98.96	32.74
44	C0	19.96	99.05	32.74
42	C1	80.04	21.03	32.74
44	C1	79.48	21.84	32.74
45	C1	79.58	21.28	32.74
42	C2	78.17	23.64	32.74
43	C2	79.80	21.39	32.74
44	C2	79.90	21.20	32.74
42	C3	77.86	23.56	12.97
43	C3	76.31	25.60	14.82
44	C3	81.22	19.17	13.62
42	C4	81.28	19.66	14.01
44	C4	80.23	20.54	14.78
45	C4	80.03	21.04	12.11

ORCID iDs

Anika Tabassum Meem  0009-0002-8444-9923

Muntasir Hossain Nadid  0009-0007-0215-2269

Md Zesun Ahmed Mia  0009-0004-3509-8455

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